

Esharaqa Afreen Jahid

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EDUCATION

Purdue University - West Lafayette

Aug 2022 – May 2026

B.S in Computer Engineering (Senior)

GPA: 3.40/4.0

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript, HTML, System Verilog, Verilog, React Native

Hardware: STM32, ESP32, FPGA, Soldering, Oscilloscope

Tools & Frameworks: GitHub, ModelSim, Cadence Virtuoso, LTspice, SolidWorks, Whisper AI, TensorFlow.js, UVM-style Verification, Altium

EXPERIENCE

Fusion Semiconductor

May 2025 - Aug 2025

Remote Hardware Design Intern

- Designed and verified a custom **SPI controller** in **SystemVerilog** as part of a reusable SoC peripheral library
- Developed **UVM-style testbenches** and automated stimulus to simulate functional correctness and edge cases; achieved 100% functional and code coverage using **ModelSim**
- Prototyped a sensor interface system on **FPGA** by connecting an I2C-based temperature sensor; implemented controller logic in Verilog and tested real-time reads using a UART debug console

Research Intern - Machine Learning in Motion

Aug 2023 – Dec 2023

Vertically Integrated Projects (VIP)

- Built a real-time 3D joint tracking system using **TensorFlow.js** for browser-based visualization.
- Created a streaming data pipeline to calculate joint angles from video input and integrated outputs into an HTML-based UI.

Ford Wilson Race of Champions

Jan 2022 – May 2022

Engineering Projects in Community Service (EPICS)

- Designed a dual-control steering system in **Fusion 360** for safe vehicle operation between a child and professional driver.
- Completed multiple chassis iterations, selected optimal electrical components, and reduced printing cost by 15%.
- Soldered components and verified circuit design using **LTspice** before hardware integration.

PROJECTS

Voice-Activated TARS (Interstellar-Inspired)

- Designed a voice assistant using **ESP32** and TFT screen with dialogue transcribed by locally installed **Whisper AI**.
- Implemented serial communication, wake-word logic, and multi-mode control (e.g., humor mode) to simulate TARS interaction.

Object-Oriented Chess Game

- Developed a polymorphic class-based engine with rule-compliant movement and threat detection logic in C++.
- Used test-driven development to write unit tests covering complex interactions and edge cases.

AHB-to-USB Interface

- Implemented pipelined SoC modules for USB TX/RX using **SystemVerilog**; verified AHB compliance and resolved RAW hazards.
- Validated functionality through waveform-driven simulation and protocol-level debug.

Rhythm Game System on STM32

- Created an interactive rhythm game using an STM32 with real-time input/output across multiple hardware modules.
- Rendered falling arrows via **TFT display (SPI)**, processed keypad input, and played audio via **DAC** and speaker.
- Implemented multiplexed **8-digit 7-segment display** that showed in game score